

AREA III: *AB Structures, Environment Engineering, Bioprocess Engineering and Allied Subjects*

1. In psychrometric chart, these represented by lines at diagonals.
 - a. Dry Bulb Temperature
 - b. Wet Bulb Temperature
 - c. Specific Enthalpy
 - d. Specific Volume

Dry Bulb Temperature - Dry bulb temperatures are represented by **vertical lines**.

Specific Enthalpy - Specific enthalpy lines run **almost parallel** to wet bulb temperature lines, but are usually labeled on the enthalpy scale.

Specific Volume - Specific volume lines are **steeper diagonals**, sloping upward to the right.

Dew Point Temperature - Found by moving **horizontally left** from a state point until reaching the 100% RH curve (saturation line).

Relative Humidity - Shown as **curved lines** from 0% (dry air) to 100% (saturated air).

Humidity ratio - Found on the **vertical axis** (y-axis).

2. This law states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.
 - a. 1st Law of Thermodynamics
 - b. 2nd Law of Thermodynamics
 - c. 3rd Law of Thermodynamics
 - d. 0th Law of Thermodynamics

1st Law - States that **energy cannot be created or destroyed**, only transformed from one form to another.

2nd law - States that **entropy of an isolated system always increases over time**, and heat flows naturally from hot to cold.

0th law - States that **if two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other**.

3. In PAES 421:2009, What is the minimum carriage way of one-lane and two-lane FMR, excluding shoulders?

- a. 2-4 m
 - b. 4-6 m
 - c. 2.5-5 m
 - d. 3-6 m

4. In PAES 402:2000, the recommended brooding temperature for 8-14 days chicks.

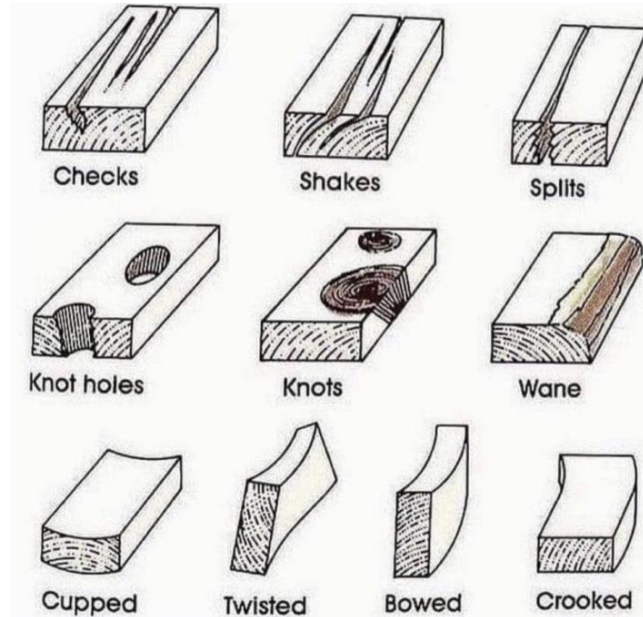
- a. provide heat when necessary
 - b. 29-32 °C
 - c. 32-35 °C
 - d. 27-29 °C

Age of chicks days	Temperature °C
1 - 7	32 - 35
8 - 14	29 - 32
14 - 21	27 - 29
Beyond 21 days	Provide heat only when necessary

5. What is the defect of wood that causes abnormal growth which occurs at the starting point of a limb or branch of a tree?

(CLARIFY WHY KNOT SHA)

- a. shake
- b. knot
- c. check
- d. wane



6. In PAES 410:2000, the lairage shall be provided with lighting intensity of _____ lux.

- a. 200
- b. 100
- c. 500
- d. 110

6.3.8 Lighting

6.3.8.1 Illumination of 110 lux shall be provided within the lairage.

6.3.8.2 For all suspect pens, lighting intensity of not less than 540 lux shall be provided.

7. What diode is needed as protection against lightning in solar installations?

- a. Bypass diode
- b. Blocking diode
- c. Shading diode
- d. Lightning diode

Bypass Diode. Protects solar cells in a module from damage due to partial shading by allowing current to bypass shaded cells.

Shading Diode. Not an official term.

Lightning Diode. Not an official term.

8. In PAES 403:2001, under the **individual nest** subsection, there shall be one nest for each ____ to ____ layer.

a. 5 – 6

c. 7 – 8

8.3.2.1 Individual nests (Figure 2a): Individual nest shall be 310 mm wide, 310 mm high and 310 mm - 360 mm long. The nest should be constructed with wire mesh at the back or with the back open for adequate air movement. The bottom of the nests shall be removable for easy cleaning. There shall be one nest for each 5 to 6 layer.

8.3.2.2 Community nest(Figure 2b): These are unpartitioned boxes about 600 mm wide and 2.4 mm long, with 200 mm opening at each end through which the layers enter and leave. Each shall have a sloping cover that is hinged so that it may be opened. The nest should be constructed with wire mesh at the back or with the back open for adequate air movement. The laying floor should be removable and should be made of 13 mm galvanized wire mesh. The laying floor shall be sloping 7% towards the other side, which is provided with foam rubber bumper pad. One nest box should be provided for 20 – 25 layers.

b. 8 – 10

d. 4 – 6

9. In CPES rating, the maximum weight rating for Environmental, Safety and Health criteria during construction.

a. 0.40

c. 0.07

b. 0.15

d. 0.30

	During Construction		Upon Completion	
	With Materials	W/o materials	With Materials	W/o materials
Workmanship	0.40	0.40	0.50	0.50
Materials	0.30	-	0.20	-
Time	0.15	0.20	0.30	0.50
Facilities	0.03	0.05	-	-
Environment, Safety and Health (ESH)	0.07	0.10	-	-
Resources Deployment	0.05	0.25	-	-

10. A complete lighting unit consisting of a lamp or lamps together with parts designed to distribute the light, to position and protect the lamps, and to connect the lamps to the power supply.

a. lumen

c. luminous

b. luminaire

d. lighting fixture

Lumen. A unit of **luminous flux**, which measures the total amount of visible light emitted by a source.

Luminous. An adjective meaning "emitting or reflecting light."

Lighting Fixture. A common term in everyday language for **luminaire**, but in technical standards (like in engineering and architecture), "luminaire" is preferred.

11. The primary law governing blackbody radiation that quantifies the intensity of radiation emitted by unit surface area into a fixed direction from the blackbody as a function of wavelength for a fixed temperature.

- a. **Stefan-Boltzmann Law**
- b. Newton's Law of Cooling
- c. Fourier's Law
- d. Planck's Law

Newton's Law of Cooling. Describes the rate of heat loss of a body to its surroundings based on temperature difference (**convection**, not radiation spectrum).

Fourier's Law. Governs **heat conduction** through materials, relating heat flux to temperature gradient.

Planck's law. Describes how the intensity and distribution of electromagnetic radiation emitted by a blackbody depend on its temperature.

12. In CPES rating, what is the percentage for time upon completion?

- a. 0.10
- b. 0.15
- c. 0.40
- d. **0.30**

13. What is the size of the wire AWG 10?

- a. 5.0 mm²
- b. **5.5 mm²**
- c. 6.0 mm²
- d. 8.8 mm²

Wire Gauge Conversion Chart

Diameter					Area				
AWG	In.	mm	CMA	mm ²	AWG	In.	mm	CMA	mm ²
4/0 (0000)	0.46	11.68	212000	107	14	0.062	1.57	4110	2.08
3/0 (000)	0.41	10.41	168000	85	15	0.057	1.45	3260	1.65
2/0 (00)	0.365	9.27	133000	67.4	16	0.051	1.30	2580	1.31
1/0 (0)	0.325	8.26	106000	53.5	17	0.045	1.14	2050	1.04
1	0.289	7.34	83700	42.4	18	0.040	1.02	1620	0.823
2	0.258	6.55	66400	33.6	19	0.036	0.91	1290	0.653
3	0.229	5.82	52600	26.7	20	0.032	0.81	1020	0.518
4	0.204	5.18	41700	21.2	21	0.0285	0.72	810	0.41
5	0.182	4.62	33100	16.8	22	0.0253	0.643	642	0.326
6	0.162	4.11	26300	13.3	23	0.0226	0.574	509	0.258
7	0.144	3.66	20800	10.5	24	0.0201	0.511	404	0.205
8	0.128	3.25	16500	8.37	25	0.0179	0.45	320	0.162
9	0.114	2.90	13100	6.63	26	0.0159	0.404	254	0.129
10	0.102	2.59	10400	5.26	27	0.0142	0.361	202	0.102
11	0.091	2.31	8230	4.17	28	0.0126	0.320	160	0.081
12	0.081	2.06	6530	3.31	29	0.0113	0.29	127	0.0642
13	0.072	1.83	5180	2.62	30	0.01	0.254	101	0.0509

AWG: American wire gauge
CMA: Circular Mils Area

14. Amount of heat added or removed measured in change of temperature without changing state.

- a. Latent Heat
- b. **Sensible Heat**
- c. Specific Heat
- d. Innate Heat

Latent Heat. Heat absorbed or released during a **phase change** (solid ↔ liquid ↔ gas) **without changing the temperature**.

Specific Heat. The amount of heat required to raise the temperature of **1 kg of a substance by 1 °C**.

Innate Heat. Not a scientific term used for heat calculations.

15. Measures the ability of the machine to remove hulls with minimum breakage.

- a. **Coefficient of wholeness**
- b. Hulling Coefficient
- c. Hulling Efficiency
- d. Fineness Modulus

3.4

coefficient of hulling

measure of the ability of the machine to remove the hulls

3.5

coefficient of wholeness

measure of the ability of the machine to remove the hulls without breaking the grain

3.8

hulling efficiency

product of the coefficient of hulling and the coefficient of wholeness of grains, expressed in percent

16. The amount paid to project owners from Clean Development Mechanism (CDM) fund for carbon emission reduction projects.

- a. **Carbon credit value**
- b. Emission value
- c. Carbon trading value
- d. Carbon emission value

17. What is the peak amperage of a solar battery charging station equipped with 16 units of 75 watt-peak 4.16-ampere solar module? The system is divided into 4 charging channels, each channel has 4 modules connected in parallel.

- a. 32.2 A
- b. 25.4 A
- c. **16.6 A**
- d. 8.3 A

18. What is the minimum methane content of biogas in order to operate biogas-fueled generators?

- a. 50%
- b. **65%**
- c. 85%
- d. 90%

19. Type of biogas plant which have aerobic conditions in the upper layers and anaerobic processes occurring in the bottom layers, especially in the settled solids?

- a. Aerobic Biogas
- b. Anaerobic Biogas
- c. **Facultative Biogas**
- d. Activated Biogas

Aerobic Biogas - Operates entirely with oxygen; no anaerobic digestion occurs.

Anaerobic Biogas - Operates entirely without oxygen; all digestion is anaerobic.

Activated Biogas - Not a standard term; may confuse with activated sludge

20. The wastewater, either treated or untreated, discharged from a reservoir, basin or treatment plant.

- a. Effluent
- b. Influent
- c. Turbid Water
- d. Hard Water

3.7

effluent

liquid waste, partially or completely treated, flowing out of a reservoir, basin, or wastewater treatment plant

3.12

influent

liquid that flows into a containing space

Hard water. Is water that contains an excess of calcium and magnesium.

21. In PAES 421:2009, the tank shall have heating device able to heat a complete charge of bituminous liquid to ____ °C.

- a. 80
- b. 120
- c. 180
- d. 200

17.3.2 Equipment

The liquid bituminous material shall be sprayed by means of a pressure distributor of not less than 1000 liters capacity, mounted on pneumatic tires of such width and number that the load produced on the road surface will not exceed 1 kN (1000kgf) per cm width of tire.

The tank shall have heating device able to heat a complete charge of bituminous liquid to 180 °C. The heating device shall be such that overheating will not occur. Consequently, the flames must not directly touch the casing of the tank containing the bituminous liquid. The liquid shall be insulated in such a way that the drop in

22. In PAES 415:2001, under natural ventilating systems of greenhouse, large vent openings provide the most ventilation with total vent area that should be ____ - ____ of floor area.

- a. 15% - 30%
- b. 30% - 50%
- c. 25% - 50%
- d. 15% - 25%

8.1 Natural ventilating system

8.1.1 Ventilation should either be built-in in the design of the roof and wall configuration.

8.1.2 There should be vent openings on both sides and ridge of a greenhouse.

8.1.3 Large vent openings provide the most ventilation, the total vent area should be 15%-25% of the floor area.

23. In PAES 421:2009, it is constructed to drain water from the subgrade and to collect surface water either from the roadway surface or adjacent.

- a. ditches
- b. culvert
- c. channel
- d. conduit

Ditches - Shallow, elongated excavations along roadsides for drainage.

23.2 Roadside Ditch

The data shall be designed on channel dimensions such as the width, depth, slope (Front slope and back slope), velocity, particle size distribution, sediment load, and discharge necessary to maintain stable channel.

This shall be constructed to drain water from the subgrade.

This shall be constructed to collect surface water either from the roadway surface or adjacent.

Culvert - Enclosed structure (usually a pipe) allowing water to pass under a road.

Channel - A broader natural or artificial waterway; not specific to road subgrade drainage.

Conduit - An enclosed pipe or passage for fluids or cables, not open drainage.

24. In PAES 408:2001, the minimum space requirement for carabao feedlot feeding shall be _____ m/animal.

- a. 0.5
- b. 1.0
- c. 0.75
- d. 1.25

Note: 0.75 is for feedlot feeding minimum space requirement. The 4 m²/animal is for the minimum space requirement of the feedlot. Do not get confused.

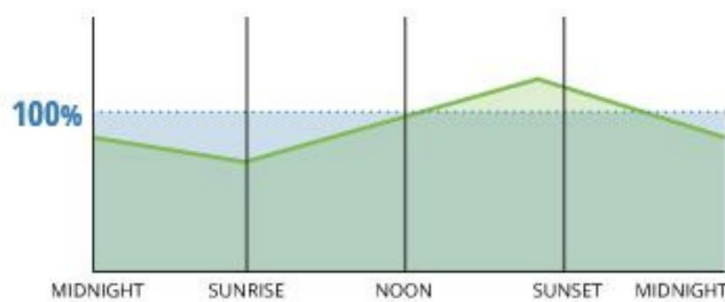
25. The maximum number of outlets for a 15-A circuit.

- a. 8-10 outlets
- b. 10-12 outlets
- c. 14-15 outlets
- d. 6-8 outlets

Note: For each 20 A circuit, no more than 10 -12 outlets is allowed.

26. Dissolved oxygen concentrations will be the lowest within a waterbody on _____.

- a. Afternoon
- b. Sunrise
- c. Sunset
- d. Noon



Note: Dissolved oxygen levels in a waterbody are lowest at sunrise because aquatic plants and algae consume oxygen during the night for respiration, and no photosynthesis occurs in the dark to replenish it.

Afternoon - DO levels are usually higher due to peak photosynthesis.

Sunrise - Lowest DO because of nighttime oxygen consumption and no oxygen production.

Sunset - DO is still relatively high from daytime photosynthesis.

Noon - DO is rising toward its peak from morning photosynthesis.

27. In PAES 401:2000, heaters shall be installed in creep areas to provide newborn pigs its required temperature of ____°C - ____°C until they are 3 days old.

- a. 32-35
- b. 29-35
- c. 30-35
- d. 27-35

8.5.1 Heaters

8.5.1.1 Heaters shall be installed in creep areas to provide newborn pigs its required temperature of 27°C - 35°C until they are 3 days old.

8.5.1.2 Heat lamps shall be placed 762 mm above the floor or 152 mm above the sow.

28. What is the mechanical power of a windmill with a diameter of 10 meters and velocity of 5 mps?

- a. 55.92 watts
- b. 981.75 watts
- c. 3,926.99 watts
- d. 9,327.56 watts

The general formula for the theoretical power of a wind is:

$$\text{Theoretical power} = (\rho A V^3) / 2 \quad \text{where: } \rho = \text{air density, } 1.25 \text{ kg/m}^3$$

$$A = \text{rotor area, m}^2$$

$$V = \text{wind speed, m/s}$$

However, what was asked is the **mechanical power** so we will be using the formula:

$$\text{Available power} = 0.245 A V^3 \quad (\text{for mechanical energy conversion, e. g. pumping})$$

For electric energy conversion, use:

$$\text{Available power} = 0.10 A V^3 \quad (\text{for electric energy conversion})$$

(CLARIFY THIS ONE KAY DILI NAKO MAKUHA ANG ANSWER IF ANG MECHANICAL ENERGY ANG GAMITON, MAKUHA SYA IF ELECTRIC ENERGY CONVERSION)

29. A Carnot engine is working between two reservoirs with temperatures of 1000°C and 500°C , what is the thermal efficiency of the system?

- a. 10%
b. 30%
c. 50%
d. 100%

Note: Since we are talking about carnot cycle, use the absolute temperature which is in Kelvin unit. And use the formula:

$$e_{max} = 1 - \frac{T_c}{T_H}$$

Efficiency maximum

Temperature of the cold reservoir

Temperature of the hot reservoir

30. Environmental-friendly refrigerant

- a. CFC
b. HFC
c. HCFC
d. FC

BY COMPARISON

Newly calculated data could help people make climate-friendly technology choices

COMPOUND	FORMULA	RADIATIVE EFFICIENCY, W/m ² -ppb	ATMOSPHERIC LIFETIME, YEARS	GLOBAL-WARMING POTENTIAL ^a
CFC-11	CCl ₃ F	0.26	45	4,660
CFC-12	CCl ₂ F ₂	0.32	100	10,200
CFC-113	CCl ₂ FCClF ₂	0.30	85	5,820
HCFC-22	CHClF ₂	0.21	11.9	1,760
HFC-134a	CH ₂ FCF ₃	0.16	13.4	1,300
HFO-1234yf	CF ₃ CF=CH ₂	0.02	0.03	<1

^a 100-year values, relative to CO₂, which has a value of 1.

31. It is the programming language of Arduino microcontrollers.

- a. R
b. Python
c. C/C++
d. Java

R - A programming language mainly for statistics and data analysis, not used for microcontrollers like Arduino.

Python - While Python can run on some microcontrollers (via *MicroPython*), Arduino's official IDE does not use Python.

Java - Java is used for Android apps and enterprise applications, not for programming Arduino boards.

32. In Arduino programming, the meaning of IDE is:

- a. Integrated Development Environment
b. Internal Development Environment

- c. International Development Environment
d. Interconnected Development Environment

33. The Feed-in-Tariff (FIT) of **solar energy** in the BiGSHOW is:

- a. 6.63 PHP/kW-hr
b. 9.68 PHP/kW-hr

- c. 5.90 PHP/kW-hr
d. 8.53 PHP/kW-hr

RE Technology	FIT Rate (Php/kWh)	Degression Rate	Installation Target (MW)
Wind	8.53	0.5% after 2 years from effectivity of FIT	200
Biomass	6.63	0.5% after 2 years from effectivity of FIT	250
Solar	9.68	5% after 1 year from effectivity of the FIT	50
Run-of-River Hydropower	5.90	0.5% after 2 years from effectivity of FIT	250

34. It is the gas produced from agricultural waste which is odorless, tasteless, colorless and **flammable**

- a. CH₄
b. CO

- c. O₂
d. CO₂

35. All are products of pyrolysis except:

- a. Vinegar
b. Tar

- c. Oil
d. Syngas

36. What is the electrical energy consumption of a 15-kW vermicast production shredder plugged in a 220-V source if it requires 2 hrs to shred 1.5 tonnes of farm waste?

- a. 0.20 kW-h/kg
b. 0.02 kW-h/kg

- c. 0.002 kW-h/kg
d. 0.022 kW-h/kg

Note: DIMENSIONAL ANALYSIS.

37. It is a type of whitening machine consisting of a ribbed cylinder enclosed in a perforated steel housing

- a. friction type
- b. abrasive type
- c. rubber roll type
- d. cone "cono" type

cone "cono" type

type of rice mill having an under-runner stone disc huller and vertical cone whitener

3.21.2

rubber roll type

type of rice mill using rubber roll huller and utilizes friction and/or combination of other types of whitener

abrasive type

type of whitening machine consisting of a cylinder or cone coated with abrasive material such as emery stone or any similar materials enclosed in a perforated steel housing

3.24.2

friction type

type of whitening machine consisting of a ribbed cylinder enclosed in a perforated steel housing

38. What is the input current to a 5 hp single phase electric motor use to drive a hammer mill at 2200 rpm. The input voltage is 220 volt. Assume a power factor of 0.98

- a. 14.5 amp
- b. 17.3 amp
- c. 20.4 amp
- d. None of the above

39. When a system deviates infinitesimally from equilibrium at every instant of its state, it is undergoing:

- a. isobaric process
- b. quasi-steady process
- c. isometric process
- d. isentropic process

Isobaric process - pressure stays constant during the process.

Isometric process (also called isochoric) - volume stays constant.

Quasi-steady process - also called *quasi-static*, the system changes so slowly that it remains infinitesimally close to equilibrium at every step.

Isentropic process - entropy stays constant (idealized adiabatic and reversible process).

40. A rice thresher that is fed with panicles at one end and discharges threshed product at the other end is called ____.

- a. radial type thresher
- b. hold-on type thresher
- c. axial-flow thresher
- d. cross-flow thresher

hold-on thresher

type of thresher wherein the panicles of the cut plants are fed into the threshing chamber while the stalks are mechanically or manually held during the threshing operation

axial flow thresher

throw-in type of thresher which allows the cut plants to move in a helical manner around the threshing cylinder with a net effect of moving the material axially between the feeding and discharge outlets

41. A wooden pole (SG=0.65), 9 cm by 9 cm by 5 m long, hangs vertically from a string in such a way that 3 m is submerged in water and 2 m is above the surface. What is the tension in the string?

- a. 19.87 N
b. 17.89 N

- c. 18.96 N
d. 16.97 N

Note: the formula to use here is the formula for buoyant force which is the **density of fluid multiplied by gravity and volume**. But by finding the tension, use the difference between the resulting forces of its weight and the buoyant force.

Find first the weight of the wooden pole by finding the density of the pole. Use the specific gravity given and multiply this with the density of water. Once the pole density is identified, you can now find the weight of the pole by multiplying the pole density, gravity, and the total volume.

$$\text{Pole density} = 0.65 (1000 \text{ kg/m}^3) = 650 \text{ kg/m}^3$$

$$\text{Weight of pole} = 650 \text{ kg/m}^3 (9.81 \text{ m/s}^2) (0.0405 \text{ m}^3) = 258.25 \text{ N}$$

Now, time to get the force of buoyancy. Still the same process but this time, use the density of water since we will be finding the force acting upon the submerged material. Also, in the volume to be used, consider only of how long the submerged part is.

$$\text{Submerged volume} = 0.09 \text{ m} (0.09 \text{ m}) (3 \text{ m}) = 0.0243 \text{ m}^3$$

$$\text{Buoyant Force} = 1000 \text{ kg/m}^3 (9.81 \text{ m/s}^2) (0.0243 \text{ m}^3) = 238.38 \text{ N}$$

Now, find the tension acting on the string by finding the difference between the Weight and buoyant force:

$$\text{Tension} = 258.25 \text{ N} - 238.38 \text{ N} = 19.87 \text{ N}$$

42. How much energy is needed for a 5-volt source to move 4 coulombs of charge?

- a. 20 joules
b. 1.25 joules

- c. 0.8 joules
d. 80 joules

Note: Since what was asked is the energy, we will be using the formula derived from the definition of voltage: the ratio of the electrical energy (Joules) to the electrical charge (Coulombs).

$$\text{Voltage} = \text{Joules} / \text{coulombs}$$

$$5 \text{ V} = E / 4 \text{ C} = 20 \text{ Joules}$$

43. It is a hypothetical activity which takes no resource or time to complete. It is represented by broken arrowed line & is used for either distinguishing activities having common starting & finishing events or to identify & maintain proper precedence relationship between activities that are not connected by events

- a. Free Float
- b. Slack Activity
- c. Dummy activity
- d. Zero Slack

Free Float → The amount of time an activity can be delayed without affecting the early start of a following activity.

Slack Activity → An activity that has extra time before it delays the project.

Zero Slack → Activities with no float; delaying them delays the whole project.

44. In lateral and sub-lateral canal where the available roadway width is 3 meters, the carriageway width shall be 2 meters and the shoulder width shall be 0.5 meters. If the available roadway width is 4 meters, the shoulder shall expand to 1 meter.

- a. road bed lateral canal
- b. roadway on main irrigation
- c. roadway on lateral & sub lateral irrigation canals
- d. Turn-out section canal

45. When you want to use objects from a drawing file in another application, you can cut these objects to the Clipboard and then paste them into another application. CUTCLIP copies selected objects to the Clipboard, removing them from the drawing using the shortcut:

- a. Ctrl Y
- b. Ctrl Z
- c. Ctrl X
- d. Ctrl V

Ctrl V = Paste → Inserts clipboard contents into the file.

Ctrl Y = Redo.

Ctrl Z = Undo.

46. In AutoCAD, it **creates copies of objects arranged in a pattern**. You can create copies of objects in a regularly spaced rectangular, polar, or path

- a. Ortho
- b. Offset
- c. Array
- d. Fillet

Ortho - Restricts cursor movement to horizontal and vertical directions (90° increments), used for drawing straight lines.

Offset - Creates a parallel or concentric copy of an object at a specified distance.

Fillet - Creates a rounded corner between two lines, arcs, or polylines.

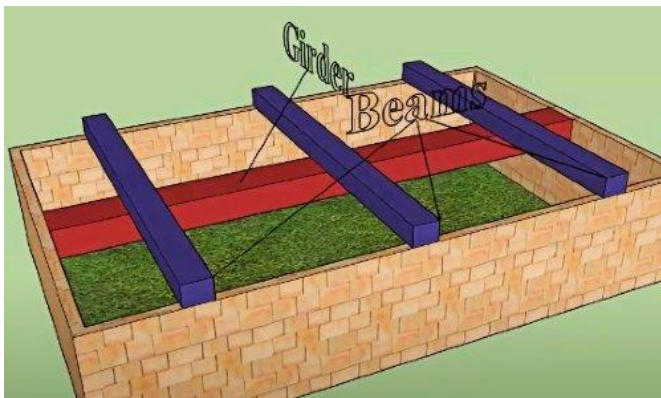
47. A structure the consists of compressive force action and supports only axial loadings is:

- a. Beam
- b. **Column**
- c. Purlins
- d. Girder

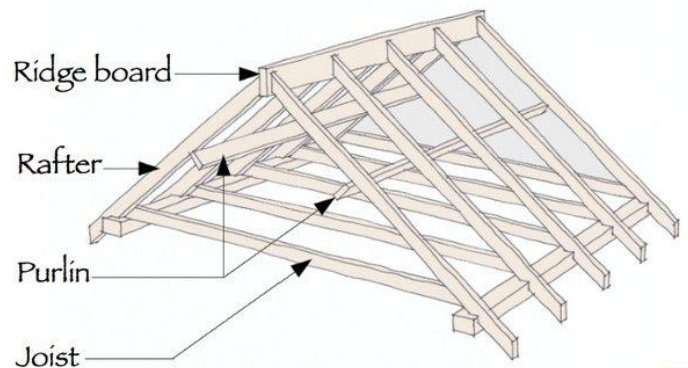
Beam - A horizontal (or slightly inclined) structural member designed to carry **bending loads** (mainly transverse to its length), not purely axial.

Purlins- Horizontal beams in roofing systems, supporting roof sheets and transferring loads to rafters or frames. They carry bending, not just axial loads.

Girder - A large beam that supports smaller beams; primarily designed for bending rather than axial compression.



PITCHED 'CUT ROOF' CONSTRUCTION



1st Associated.co.uk ©

48. A beam needs 4 pcs of 20 mm RSB. If there is no available 20 mm RSB, how many 16 mm RSB can substitute for 20 mm?

- a. 4 pcs
- b. 5 pcs
- c. 6 pcs
- d. 7 pcs

Note: in finding how many RSB is needed, you'll just have to find the cross sectional area of the different sizes of RSB mentioned. In this case, we will be basing our answer on the total CSA of 20 mm as to how many pieces is needed when using a 16 mm RSB.

$$\text{TOTAL CSA OF 20 mm} = 4 \text{ pcs } \left(\frac{\pi(20\text{mm})^2}{4} \right) = 1256.64 \text{ mm}^2$$

$$\text{CSA OF 16 mm} = \left(\frac{\pi(16\text{mm})^2}{4} \right) = 201.06 \text{ mm}^2$$

$$\text{Number of 16 mm bars} = 1256.64 / 201.06 = 6.25 \text{ so round it up to 7 pcs.}$$

49. How many pieces of umbrella nails are there if there are 10 kilograms of umbrella nails?

- a. 1,000 pcs
- b. 1,100 pcs
- c. 1200 pcs
- d. 1250 pcs

Note: in every type of nail material, there is a designated number of nails per kilogram. See photo below:

Materials	Number per kilogram
GI Roof Nails	120
GI Rivets	180
GI Washers	126
Lead Washers	75
Umbrella Nails	120

So since in the problem, an umbrella nail was asked, the answer should be 1200 pcs in 10 kilogram.

50. It a positively charged particle made up of **two protons and two neutrons** and released by a radioactive nucleus

- a. **Alpha radiation**
- b. Beta radiation
- c. Ultra-violet radiation
- d. Gamma radiation

51. Given the initial temperature of 45 F and initial pressure of 16 bar, the pressure changed to 35 bar, determine the change in temperature.

- a. 250
- b. 300
- c. 500
- d. **600**

Note: use the gas law. But first, convert the temperature to rankine .

GAS LAW	FORMULA
Boyle's law	$P_1 V_1 = P_2 V_2$
Charles's law	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$
Gay-lussac's law	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
Combined law	$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$
Ideal gas law	$PV = nRT$

52. It is the production of food, fiber, or other plant or animal products using farming techniques that protect the environment, public health, human communities, and animal welfare.

- a. Organic Agriculture
- b. **Sustainable Agriculture**
- c. Traditional Agriculture
- d. Conventional Agriculture

Organic Agriculture - Farming without synthetic fertilizers, pesticides, or GMOs; relies on natural inputs and processes.

Traditional Agriculture - Farming methods passed down through generations, often low-tech and culturally rooted.

Conventional Agriculture - Modern, high-input farming using synthetic fertilizers, pesticides, and machinery for maximum yield.

53. It is a production system that sustains the health of soils, ecosystems and people. It relies on ecological processes, biodiversity and cycles adapted to local conditions, rather than the use of inputs with adverse effects.

- a. **Organic Agriculture**
- b. Sustainable Agriculture
- c. Traditional Agriculture
- d. Conventional Agriculture

54. It is the layer of aggregate, soil-treated aggregate, treated soil, or soil aggregate that rests upon the Subbase or if no Subbase, upon the sub-grade. Treatment may include application of chemical-based soil additives such as soil-stabilizers and/or any approved method.

- a. Aggregate Base Course
- b. **Base Course**
- c. Subbase Course
- d. Aggregate Subbase Course

4.3 **base course**

the layer of aggregate, soil-treated aggregate, treated soil, or soil aggregate that rests upon the Subbase or if no Subbase, upon the sub-grade. Treatment may include application of chemical-based soil additives such as soil-stabilizers and/or any approved method

4.39 **Subbase course**

refers to the layer of the specified or selected materials of designated thickness in a pavement structure immediately above the sub-grade and below the base course



55. It involves the arrangement of compost mix in long, narrow piles or windrows that are periodically turned to maintain aerobic conditions

- a. Aerated static pile
- b. In-vessel system composting
- c. **Windrow composting**
- d. Bin composting

Aerated static pile - Composting in large piles with forced aeration (using blowers or pipes), no turning needed.

In-vessel system composting - Composting inside enclosed containers or drums with controlled temperature and aeration.

Bin composting - Composting in bins or small containers, common for small-scale/home use.

56. The percent of moisture content of test materials for peanut sheller.

- a. 8%-10%, wet basis
- b. 12%-14%, wet basis
- c. 8%-10%, dry basis
- d. 12%-14%, dry basis

Note: This is based on PAES 221: 2004. However, for further understanding, we use wet basis because it is more practical, standard and easier to interpret.

Test materials to be used shall have the following characteristics:

4.6.1.1 Variety : commonly or locally grown

4.6.1.2 Moisture content : 12 % - 14 %, wet basis

57. **Measures velocity by the change of source light or sound frequency** from the frequency of reflections from moving particles.

- a. Optical Strobe
- b. Doppler
- c. Bellow
- d. Platometer

Optical Strobe - uses a flashing light to measure rotational speed or RPM.

Bellow - a device used in fluid pressure measurement or mechanical expansion systems and not for velocity.

Platometer - also known as planimeter. used for measuring flatness or thickness of materials and not velocity.

58. Ratio of the weight of corn kernels that fell out from the machine during shelling operation to the weight of the total corn kernel input of the sheller, expressed in percent.

- a. Scattering loss
- b. Cylinder loss - *JUST MADE UP TERM*
- c. Separation loss
- d. Blower loss

Note: This is a standard from PAES 208: 2000.

3.1

blower loss

ratio of the weight of corn kernels blown by the sheller fan, to the weight of the total corn kernel input of the sheller, expressed in percent

3.20

scattering loss

ratio of the weight of corn kernels that fell out from the machine during shelling operation to the weight of the total corn kernel input of the sheller, expressed in percent

3.21

separation loss

ratio of the weight of corn kernels that come out of the shelling chamber with the cobs at the cob outlet, to the weight of the total corn kernel input of the sheller, expressed in percent

3.29

unshelled loss

ratio of the weight of corn kernels that remained in the cobs of the corn fed into the shelling chamber, to the weight of the total corn kernel input of the sheller, expressed in percent

59. A 2.4-tonne capacity *Jatropha* oil expeller was loaded full with dried *Jatropha* seeds in 4 minutes. It produced 600 kg of crude oil in 2 hours. What is its crude oil production rate?

- a. 300 kg/h
b. 200 kg/min
c. 600 kg/h
d. 600 kg/min

Note: Just divide the produced crude oil with two since it was stated in the problem that in two hours, it will produce 600 kg and that will be the value of production rate kg per hour.

60. A 10MW-25MW hydro power plant is classified as: _____.

- a. Micro hydro
b. Small hydro
c. Mini hydro
d. Medium hydro

Small hydro power uses the flow of water to turn turbines that are connected to a generator for the production of electricity. Small hydro is divided into further categories depending on its size, such as mini- (less than 1000kW), micro- (less than 100kW) and pico-hydro (less than 5kW) (EHSa 2005); the definition may vary according to manufacturers and countries, as there is no internationally accepted definition of small hydropower. In China, small hydropower refers to capacities of up to 25 MW, in India of up to 15 MW and in Sweden 'small' refers up to 1.5 MW. However, a capacity of up to 10 MW is a generally accepted one. The European Small Hydropower Association (ESHA), the European Commission and UNPEDE (International Union of Producers and Distributors of Electricity

Reference: <https://www.ctc-n.org/technologies/small-hydropower>

Hydro plants are classified based on their capacities, as follows:

- (i) micro-hydro - 1 to 100 kW;
- (ii) mini-hydro - 101 kW to 10 MW; and
- (iii) large hydro - more than 10 MW.

Reference: <https://legacy.doe.gov.ph/hydropower>

61. Convert the binary number 010101_2 to decimal number system

- a. 53_{10}
b. 85_{10}
c. 42_{10}
d. 21_{10}

Note: This is easy by using your scientific calculator. Go to Mode - Base - enter given - choose BIN - equal - choose DEC. This process goes the same way for other base, just press the right button in the calculator.

62. Which one is an integrated circuit used in instrumentation and control?

- a. TD 8062
b. LM333
c. ES8813
d. CQ 556

TD 8062 - not a standard IC used in instrumentation (no widely known reference).

ES8813 - an audio codec IC, not typically for instrumentation/control.

CQ 556 - mainly used in timing circuits, not primarily for instrumentation.

63. In PAES 421:2009, the minimum radius of curvature for flat, rolling and mountainous terrains respectively shall be.
- a. 75, 50 & 5 m
 - b. 500, 150 & 50 m
 - c. 150, 100 & 50 m
 - d. 600, 200 & 100 m

8.1 Design speed

There shall be design speed provisions to lessen the occurrence of accidents due to over speeding.

Radius of Curvature

The minimum radius design for curvature shall be 500 meters for flat terrain, 150 meters for rolling terrain and 50 meters for mountainous terrain. Avoid minimum radius at the beginning and end of all straight sections with a length of more than 5 kilometers and or isolated curves. Curves with larger radius must be used where terrain permits.

64. It refers to the extent or degree by which the bran layer of the brown rice is removed as a result of whitening
- a. Whitening degree
 - b. Milling degree
 - c. Milling recovery
 - d. Milling index

3.11

milling capacity

quantity of paddy that the rice mill can process to a specified quality per total milling time, expressed in metric tons per hour

3.12

milling degree

extent or degree by which the bran layer of the brown rice is removed as a result of whitening

3.13

milling recovery

ratio of the weight of milled rice to the total weight of paddy, expressed in percent

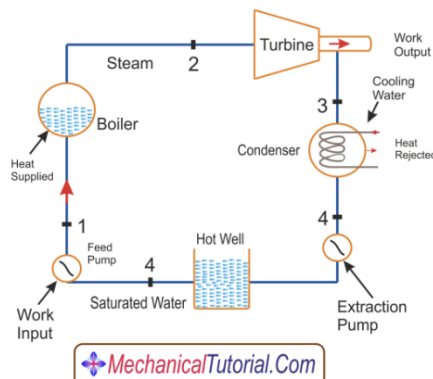
3.14

milling recovery index

ratio of the milling recovery obtained in actual testing, to the milling recovery obtained from the laboratory test mill

65. The most efficient and ideal gas cycle.
- a. Rankine Cycle
 - b. Carnot Cycle
 - c. Otto Cycle
 - d. Brayton Cycle

Rankine Cycle - is the fundamental operating cycle of all powerplants where an operating fluid is continuously evaporated and condensed. A theoretical cycle that converts heat into mechanical energy working at a constant pressure.



Otto Cycle - this is a thermo cycle which explains how internal combustion engines convert chemical energy from gasoline into motion.

Brayton cycle - is a power conversion system that uses a gas to produce thrust in jet engines or power in natural gas turbines by compressing and heating air.

66. Standard thickness of concrete of trapezoidal ditch w/ riprap
- a. 0.25 m
 - b. 0.15 m
 - c. 0.30 m
 - d. 0.20 m

Construction Requirements

Trapezoidal ditch design shall be constructed with grouted ripraps or with concrete lining.

It shall have a minimum thickness of 0.25 meters.

67. Lighting computation in a warehouse with 4x4 grid lighting layout with mounting height ratio of 1.0
- a. 4 m
 - b. 16 m
 - c. 2 m
 - d. 8 m

Note: this is just a ratio of the spacing to mounting height.

68. Contains less than or equal to 20% longitudinal streaks
- a. Well-milled rice
 - b. Regular milled rice
 - c. Overmilled rice
 - d. Undermilled rice

3.9

regular milled rice

rice grain from which the hull, the germ, the outer bran layers and the greater part of the inner bran layers have been removed but part of the lengthwise streaks of the bran layers may still be present on 21% to 40% of the sample grains

3.10

undermilled rice

rice grain from which the hull, the germ, the outer bran layer, and the greater part of the inner bran layer have been removed, but part of the lengthwise streaks of the bran layer may still be present on more than 40 % of the sample grains

3.11

well-milled rice

rice grain from which the hull, the germ, the outer bran layer, and the greater part of the inner bran layer have been removed, but part of the lengthwise streaks of the bran layer may still be present on 1% to 20% of the sample grains

69. Moisture gradient of corn and rice in mechanical dryer
- a. 0.5%
 - b. 1.0%
 - c. 2.5%
 - d. 2.0%

Table 1 – Performance Criteria for Mechanical Grain Dryer

CRITERIA	Performance Data	
	Rice	Corn
1. Moisture gradient, %, max.		
- Batch type	4.0	4.0
- Recirculating type	2.0	2.0

70. It is the chamber maintained under pressure for uniform distribution of the heated air through the grain mass
- dust collection system
 - drying chamber
 - plenum
 - tempering bin

Dust collection system - Removes dust and fine particles during processing, not for distributing heated air.

Drying chamber - The main space where grains are dried, but air distribution isn't controlled here.

Tempering bin - Used after drying to allow moisture redistribution in the grain, not for air distribution.

71. The rate of radiation of a black body at 0°C is E watt. Then the rate of radiation of this black body at 273°C will be:
- $8E$
 - $16E$
 - $4E$
 - E

Note: use Stefan Boltzmann Law but convert the temperature first into kelvin as an absolute temperature and find the ratio of the 2nd temp to the 1st temp. • $E = \sigma T^4$

72. in PAES 420:2002, the door height shall be provided with a clearance of at least ____ m above the tallest machinery
- 0.3 m
 - 0.4 m
 - 0.6 m
 - 0.7 m

6.4 Door/Gate

6.4.1 Minimum dimension of a doorway shall be 3 m high and 3.7 m wide. For bigger equipment, a clearance of 0.6 m on both sides shall be added.

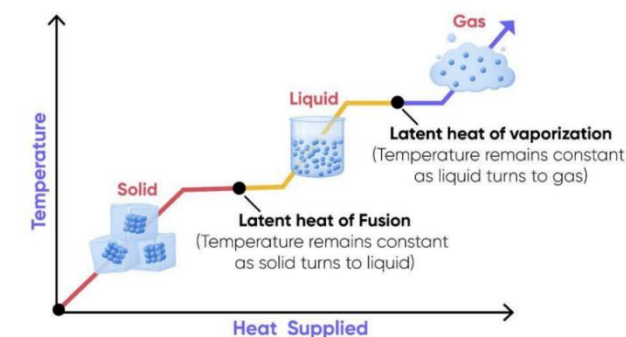
6.4.2 The door height shall be provided with a clearance of at least 0.3 m above the tallest machinery.

6.4.3 Overhead or rolling doors are recommended to be used than the hinged doors.

6.4.4 There shall be provision for a smaller service door so that the large door does not have to be opened for routine activities.

73. Heat absorbed by a unit mass of a material at its boiling point to convert the material into a gas without temperature change
- Latent heat of sublimation
 - Latent heat of deposition
 - Latent heat of fusion
 - Latent heat of Vaporization

LATENT HEAT



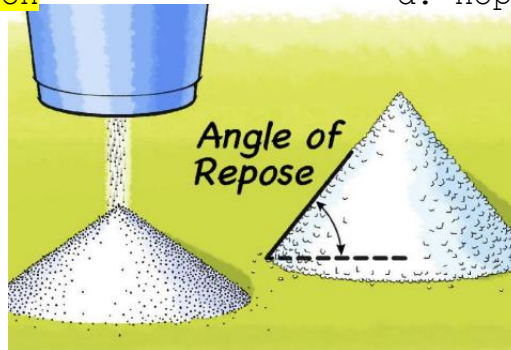
74. Which of the following is a good insulator

- a. Brass
- b. Mica
- c. Tin
- d. Platinum

All are metals which are good conductors except Mica.

75. The angle measured from the horizontal at which product will start to **move downwards over a smooth surface with gravity discharging** the product is called a.

- a. angle of discharge
- b. angle of friction
- c. angle of repose
- d. hopper angle



76. In PAES 616:2016, precautionary measures in *Wastewater Re-Use for Irrigation*, which is false?

- a. Direct contact with wastewater shall be avoided
- b. Use of fine mist for sprinkler irrigation shall be used
- c. Irrigation with wastewater shall be stopped immediately when algal bloom occurs
- d. Potable and wastewater lines shall not cross connect

10 Precautionary Measures

10.1 Direct contact with wastewater shall be avoided.

10.2 Use of fine mist for sprinkler irrigation shall be avoided to minimize the risk of aerosol dispersion by wind drift.

10.3 Potable and wastewater lines shall not cross-connect. Wastewater pipeline shall be installed far enough from a parallel potable water pipeline.

10.4 Storage facilities shall be designed such that seepage is prevented and freeboard is adequate.

10.5 Irrigation with wastewater shall be stopped immediately when algal bloom occurs.

77. Corn Grits with 0.86 mm to 1.09 mm length.

- a. Grit #16
b. Grit #18
c. Grit #10
d. Grit #14

3.5.1

grit #10

milled corn kernels with particle size between 1.8 mm to 2.0 mm

3.5.2

grit # 12

milled corn kernels with particle size between 1.5 mm to 1.7 mm

3.5.3

grit # 14

milled corn kernels with particle size between 1.2 mm to 1.4 mm

3.5.4

grit # 16

milled corn kernels with particle size between 1.10 mm to 1.19 mm

3.5.5

grit # 18

milled corn kernels with particle size between 0.86 mm to 1.09 mm

78. The amount of moisture in the grain expressed as a percentage of the total weight of the samples.

- a. Moisture content dry basis
b. Moisture content wet basis
c. Moisture content
d. Equilibrium moisture Content

Moisture content dry basis - % water relative to dry matter only, not total weight.

Moisture content - incomplete; must specify wet or dry basis.

Equilibrium Moisture Content (EMC) - the moisture content where the grain is in balance with surrounding air (no gain/loss)

79. Recommended **stacking density for corn in terms of number of bags** per cubic meter.

- a. 10
b. 9
c. 12
d. 15

Table 2 – Recommended stack heights

Type of grain	Maximum stack height inlayers	Stack height, m
Wheat, barley and corn	18	4.57
Paddy	16	6
Rice	16	5.5

Table 3 – Recommended stacking density

Stacking Density	Number of bags per cubic meter
Paddy	10
Rice	15
Corn	12

80. A 1m x 100m poultry handling area is to be lighted at an average maintained footcandle of 50 fc. How many 2-lamp fixtures of 120cm long F40 T12 WW rapid start fluorescent lamps are required assuming the CU is 0.35 and the LLF is 0.75?

- a. 3 fixtures
- b. 6 fixtures
- c. 2 fixtures
- d. 4 fixtures

Note: there is an error of unit here, 50 fc should be 50 lux since we need to convert the footcandle to lux in order to have a fixed unit analysis here and if we convert 50 fc to lux, it will be 538.2 lux which will give us a higher total lumens required which may not coincide to the choices when fixtures is solved.

Formula to use:

$$\text{Total \# of fixtures} = \frac{\text{Illumination (Area)}}{\text{Lamp per fixture (lumens)(cu)(mf)}}$$

Total lumens required = illumination (Area)

$$= 50 \text{ lux } (100 \times 1\text{m}) = 5000 \text{ lumens}$$

Lamp Output = 3200 lumens based on 120 cm F40 T12 WW fluorescent light but there are 2 lamp-fixtures so it is 6400 lumens in total. Note: there are fixed number of lumens on specified lamps like F40 T12 WW. (ASK FOR A REFERENCE FOR THIS!)

$$\text{Effec. useful lumens/fixture} = 6400 \text{ lumens } (0.35)(0.75) = 1680 \text{ lm}$$

$$\text{Number of fixtures} = 5000/1680 = 2.98 \sim 3 \text{ fixtures}$$

81. The heat exchange when a material changes its state of molecular aggregation without change in temperature.

- a. Latent heat
- b. Specific heat
- c. Sensible heat
- d. All of the above

82. A systematic approach to identifying, evaluating, and controlling food safety hazards.

- a. CPES
- b. HACCP
- c. GMP
- d. SSOP

CPES - a system of grading performance of a constructor for a project using a set of criteria.

GMP - a system that ensures that manufactured products - are consistently produced and controlled according to set quality standards.

SSOP - a written specific document of procedures or programs used to maintain equipment and the environment in a sanitary condition for food processing. It is a step - by - step description of cleaning.

83. For **the purely inductive circuit**, the power factor is:

- a. **0** c. >1
b. <1 d. 1

<1 – true for most **reactive circuits**, but for **purely inductive** it's specifically **0**.

>1 – impossible; PF cannot exceed 1.

1 – unity PF only for purely resistive loads ($\phi = 0^\circ$).

84. What force is required to punch a 20 mm diameter hole in a plate that is 25 mm thick? The shear strength is 350 MPa

- a. **550 kN** c. 500 kN
b. 400 kN d. 450 kN

85. In PAES 418:2002, area in sqm per ton of commodity in primary processing plant is:

- a. 10 sqm per ton c. **20 sqm per ton**
b. 15 sqm per ton d. 25 sqm per ton

86. When the surface of an air-cooling evaporator falls below 0°C , frost will form. Frost is:

- a. Air and ice c. Frozen water
b. **Air and water** d. Ice foam

87. A truck with axle loads of 40 kN and 60 kN on a wheel base of 5 m rolls across a 10-m span. Compute the maximum bending moment.

- a. **160 kN·m** – **NEED TO WORK ON MY SOLVING** c. 245 kN·m
b. 122.5 kN·m d. 320 kN·m

88. Biochemical Oxygen Demand (BOD_5) in mg/L for wastewater re-use:

- a. <500 c. **<150**
b. <200 d. <50

Table 1. Limits on wastewater quality for irrigation

Parameter	Unit	Landscape Irrigation	Method of Analysis	Crop Irrigation		
				Food eaten raw and not commercially processed	Food crops commercially processed	Non-food crops
For crop productivity and protection of environment						
Bicarbonates ¹	mg/L	<500		<500	<500	<500
Biochemical Oxygen Demand (BOD ₅)	mg/L	<150	Azide Modification (Dilution Technique)	<150	<150	<150
Electrical Conductivity	µS/cm	<2000		<1000	<1000	<2000
Free Residual Chlorine	mg/L	<1		<1	<1	<1
pH		6.5 – 8.0	Glass Electrode Method	6.5 – 8.0	6.5 – 8.0	6.5 – 8.0
Sodium Adsorption Ratio (SAR)		<18		<18	<18	<18
Sodium (Na)	meq/L	<3		<3	<3	<3
Total Nitrogen (TN) ¹	mg/L	<30		<30	<30	<30
Total Phosphorus (TP) ¹	mg/L	<30		<30	<30	<30
Total Suspended Solids	mg/L	<140	Gravimetric Method	<140	<140	<140
For protection of animal and human health						
Ascaris	MPN/100 mL	0		0	0	0
Fecal Coliform	MPN/100 mL	<200	Multiple-Tube Fermentation Technique or Membrane Filter	Not detectable ²	<200	<200
Nematodes	MPN/100 mL	0		0	0	0

¹ None to moderate degree of restriction

² Not detectable – means the total number of total fecal coliform organisms shall not exceed 14 MPN/100 ML in any sample

ADOPTED AND MODIFIED from Department of Agriculture. 2007. Guidelines on the Procedures and Technical Requirements for the Issuance of a Certification Allowing the Safe Re-Use Of Wastewater for Purposes of Irrigation and Other Agricultural Uses and DENR Administrative Order 35 Series Of 1990 – Revised Effluent Regulations.

89. In an inductive circuit, the inductance is 1.5H. What is the inductive resistance at 50Hz source?

- a. 471 ohms
b. 416 ohms
c. 461 ohms
d. 417 ohms

Note: We were asked to find the inductive resistance that is also known as inductive reactance, therefore, we will be using the formula $X_L = 2\pi fL$ wherein, f is the frequency of the AC in hertz and L is the inductance of the inductor in Henries.

90. Corn initially at 24% is to be dried to 14%. If the initial weight of the shelled corn is 1000 kilos, what is its final weight?

- a. 893.72 kg
b. 783.72 kg
c. 883.72 kg
d. 383.72 kg

Note: in finding in either its initial or final weight, you use this formula:

Weight loss during drying

During drying, paddy grain will loose weight due to loss of moisture:

$$W_f = W_i \times \frac{100 - MC_i}{100 - MC_f}$$

W_i = Initial weight [g]

W_f = Final weight [g]

91. In PAES 411:2000, the gut room/tripery shall be provided with lighting intensity of _____ lux.

- a. 200
b. 220
c. 540
d. 110

92. Final weight for 9 tons of rice to be dried from 24% to 14%.

- a. 7.9 ton
b. 8.9 ton
c. 9.9 tons
d. 10.9 tons

93. In PAES 409:2001, how many milking stalls are required for a milking herd having 21-30 animals with a milking machine?

- a. 1
b. 2
c. 3
d. 4

Table 1 – Recommended number of milking stalls, machines and operators

Animals	No. of milking stalls	No. of operator		No. of milking machines* (if used)
		Without milking machine	With milking machine	
1-10	1	1	1	1
11-20	2	2	2	1
21-30	3	3	2	1

* Two buckets

94. Determine the volume of the digester required to produce biogas for a livestock farm having 500 heads of cattle. The projected retention time is 25 days with feed material to water ratio of 1:1.5.

a. 445 m³

c. 424 m³

b. 469 m³

d. 438 m³

95. What is the equivalent size in mm² of a 250 MCM cable?

a. 126.67 mm²

c. 146.67 mm²

b. 136.67 mm²

d. 156.67 mm²

96. Burning efficiency of the biomass furnace shall be at least:

a. 98%

c. 90%

b. 95%

d. 80%

6.4 Heating system efficiency of the furnace shall be at least 65% for direct-fired and 50% for indirect-fired. (PAES 201:2000)

6.5 Burning efficiency of the biomass furnace shall be at least 95%.

6.6 Furnace efficiency shall be at least 65%.

97. Stoichiometric air requirement for pyrolysis

a. 10-20%

c. 30-40%

b. 20-30%

d. 40-50%

98. The C/N Ratio of agricultural waste

a. 10:20

c. 30:40

b. 20:30

d. 40:50

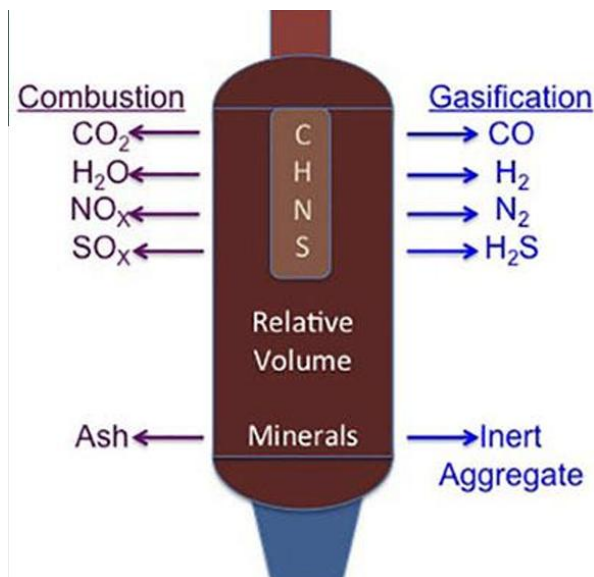
99. Main product of gasification

a. CH₄

c. O₂

b. CO

d. CO₂



100. The heating system efficiency for rice hull direct-fired mechanical grain dryer

a. 95%

c. 90%

b. 80%

d. 65%

Table 1 – Performance Criteria for Mechanical Grain Dryer

CRITERIA	Performance Data	
	Rice	Corn
1. Moisture gradient, %, max.		
- Batch type	4.0	4.0
- Recirculating type	2.0	2.0
2. Drying rate, %/h	0.8 (Batch Type) 1.0 (Continuous flow)	2.0 (Shelled Corn) 0.4 (Ear Corn)
3. Product quality*		
- Cracked grain, % increase, max.	3.0 (Batch Type) 2.0 (Continuous flow)	10
- Head rice, % decrease, max.	5.0	N/A
- Hulled/damaged grain, % increase, max.	2.0	N/A
- Broken/Split kernels, % increase, max.	N/A	7
4. Drying Loss/Spillage, %, max.	1.0	1.0
5. Drying efficiency, % min.	75	75
6. Heating system efficiency, % min		
- Petroleum based fuel direct-fired	90	90
indirect-fired	75	75
- Biomass fuel direct-fired	65	65
indirect-fired	50	50
7. Blower flow rate, cmm/kg (cfm/ton)	0.05 (1750)	0.05 (1750)
8. Static pressure, mm H ₂ O (in.H ₂ O)	25 (1)	25 (1)

* allowable difference between the laboratory analysis and machine's performance

NAME: R.L.A.

SCORE: _____